Lab 8

Goals:

1. Implement and understand Banker's Algorithm
   - Read over the slides in Bankers.pdf
   - Try to implement the algorithm described
   - Try out the example problems below

a. Compute the Need table/matrix. If there is a safe sequence find one.

Total instances: A = 7, B = 3, C = 6

Available

A	B	C
3	1	2

Allocation

	A	B	C
T0	1	0	1
T1	2	1	0
T2	1	1	1
T3	0	0	2

Max

	A	B	C
T0	3	2	2
T1	2	2	2
T2	4	2	2
T3	2	1	3

b. Compute the total instance for A, B, C.

Compute the Need table/matrix.

Can the request T2 (1,0,1) be done/accepted?

If there is a safe sequence find one.

Available

A	B	C
2	1	0

Allocation

	A	B	C
T0	1	0	0
T1	1	1	0
T2	1	0	1
T3	0	1	1

Max

	A	B	C
T0	3	2	2
T1	1	2	1
T2	3	1	2
T3	1	1	2

c. Compute the Need table/matrix.

Can the request T1 (1,0,2) be done/accepted?

If there is a safe sequence find one.

Available

A	B	C
3	3	2

Allocation

	A	B	C
T0	0	1	0
T1	2	0	0
T2	3	0	2
T3	2	1	1
T4	0	0	2

Max

	A	B	C
T0	7	5	3
T1	3	2	2
T2	9	0	2
T3	2	2	2
T4	4	3	3

d. Compute the Need table/matrix.
If there is a safe sequence find one.

Available

A	B	C
1	1	1

Allocation

	A	B	C
T0	0	1	0
T1	2	0	0
T2	3	0	3
T3	2	1	1
T4	0	0	2

Max

	A	B	C
T0	7	5	3
T1	3	2	2
T2	9	0	4
T3	2	2	2
T4	4	3	3

2. Implement and understand the Detection Algorithm

- Read over the slides in detection.pdf
- Try to implement the algorithm described
- Try out the example problems below

a. Apply the detection algorithm. List any processes that get stuck in deadlock, or list one possible sequence they can finish in.

Available

A	B	C
1	2	1

Allocation

	A	B	C
T0	1	0	0
T1	0	1	1
T2	1	1	0
T3	0	0	1

Request

	A	B	C
T0	0	1	0
T1	1	0	0
T2	1	0	1
T3	0	0	1

b. Apply the detection algorithm. List any processes that get stuck in deadlock, or list one possible sequence they can finish in.

Available

A	B	C
0	1	1

Allocation

	A	B	C
T0	0	1	0
T1	2	0	1
T2	1	1	0
T3	0	0	2

Request

	A	B	C
T0	0	0	1
T1	1	0	0
T2	0	1	1
T3	0	0	1

c. Apply the detection algorithm. List any processes that get stuck in deadlock, or list one possible sequence they can finish in.

Available

A	B	C
1	0	1

Allocation

	A	B	C
T0	0	1	0
T1	1	0	1
T2	1	1	0
T3	0	0	2
T4	1	0	0

Request

	A	B	C
T0	0	0	1
T1	0	1	0
T2	1	0	0
T3	0	0	1
T4	0	1	1

d. Apply the detection algorithm. List any processes that get stuck in deadlock, or list one possible sequence they can finish in.

Available

A	B	C
0	0	0

Allocation

	A	B	C
T0	0	1	0
T1	2	0	0
T2	3	0	3
T3	2	1	1
T4	0	0	2

Request

	A	B	C
T0	0	0	0
T1	2	0	2
T2	0	0	1
T3	1	0	0
T4	0	0	2

e. Apply the detection algorithm. List any processes that get stuck in deadlock, or list one possible sequence they can finish in.

Available

A	B	C
1	0	0

Allocation

	A	B	C
T0	1	0	0
T1	0	1	0
T2	1	1	0
T3	0	0	1

Request

	A	B	C
T0	0	0	0
T1	0	0	2
T2	1	0	0
T3	0	2	0